

Experiment 'p_6_1_d_process_array_first_and_last_seq'

Results

June 4, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing assume : 0
Responses containing expect : 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_6_1_d_process_array_first_and_last_seq

Natural language statement: Write a method that takes an array of ten integers and then returns the first and last elements.

Method signature: p_6_1_d_process_array_first_and_last_seq(inputs: seq<int>) returns (first_and_last: seq<int>)

Ensures

- `first_and_last == [inputs[0], inputs[|inputs| - 1]]`

Requires

- `|inputs| == 10`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes an array of ten integers and then returns the first and last
↪ elements.

The signature should be:

p_6_1_d_process_array_first_and_last_seq(inputs: seq<int>) returns (first_and_last: seq<
↪ int>)

The method should respect the following contract:

requires |inputs| == 10, ensures first_and_last == [inputs[0], inputs[|inputs| - 1]]

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn0A91GGMgKJ5s6Uw3DuR2u3S7wsr

Model: gpt-5.5-2026-04-23

Created at: 1780571069

Finish reason: stop

Usage: 322 tokens in, and 122 tokens out

```
{  
  first_and_last := [inputs[0], inputs[|inputs| - 1]];  
}
```

Final Program

```
method p_6_1_d_process_array_first_and_last_seq(inputs: seq<int>) returns (first_and_last  
↪ : seq<int>)  
  requires |inputs| == 10  
  ensures first_and_last == [inputs[0], inputs[|inputs| - 1]]  
{  
  first_and_last := [inputs[0], inputs[|inputs| - 1]];  
}
```

Total Token Usage

Input tokens: 322

Output tokens: 122

Reasoning tokens: 81

Sum of ‘total tokens’: 444

Experiment Timings

Overall Experiment started at 1780571068589, ended at 1780571088925, lasting 20336ms (20.34 seconds)

Iteration #1 started at 1780571068611, ended at 1780571088925, lasting 20314ms (20.31 seconds)